

Name:

Date:

Standards: MA.5.GR.3.1-3.3

Answer each question. Show your work when asked.

This check covers the lessons listed above.

1

A rectangular prism is 5 cubes long, 3 cubes wide, and 4 cubes tall. What is the volume?

2

Find the volume of a cuboid: length 9 cm, width 6 cm, height 4 cm.

3A cuboid has a volume of 180 cm^3 , a length of 9 cm, and a width of 5 cm. Find the height.**4**

A cube has a volume of 64 cubic inches. What is the length of one edge?

5

Classify: ■■■a) $3,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$ ■■■b) $750 \text{ mL} = \underline{\hspace{2cm}} \text{ cm}^3$ ■■■c) $2 \text{ L} = \underline{\hspace{2cm}} \text{ mL}$

6

Two cuboids are joined: A is $8 \text{ cm} \times 4 \text{ cm} \times 3 \text{ cm}$ and B is $5 \text{ cm} \times 4 \text{ cm} \times 2 \text{ cm}$. What is the total volume?

7

A tank is $25 \text{ cm} \times 20 \text{ cm} \times 30 \text{ cm}$. How many liters can it hold?

8

A cuboid tank ($30 \text{ cm} \times 20 \text{ cm} \times 15 \text{ cm}$) is half full of water. How many milliliters of water are in the tank?

9

An L-shaped solid has outer dimensions $12\text{ cm} \times 8\text{ cm} \times 5\text{ cm}$ with a $4\text{ cm} \times 3\text{ cm} \times 5\text{ cm}$ piece removed. Find the volume.

10

A rectangular prism with volume 100 cm^3 must have edges of 10 cm, 5 cm, and 2 cm. Is Kai correct? Explain why or why not, and give another set of whole-number dimensions.